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Chapter 1

Fourier Analysis

1.1 Introduction

Complex exponentials are functions of particular interest to mathematicians, scientists, and engineers. Recall the following:

$$\frac{d}{dt}e^{st} = se^{st} \tag{1.1}$$

$$\int e^{st} dt = \frac{1}{s}e^{st} + C \tag{1.2}$$

where s is a complex constant.

The complex exponential e^{st} is an eigenfunction of differentiation with eigenvalue s . Because of this fact, complex exponentials are solutions to many differential equations and generate simple input-output relationships for systems described by differential equations.

The preceding commentary might suggest that systems described by differential equations could be simply analyzed by decomposing inputs into a sum of complex exponentials. This possibility was first explored by Joseph Fourier in 1821 as he used complex exponentials to solve problems involving heat flow. This type of analysis, called Fourier analysis in his honor, is used today in diverse fields to great effect.

Chapter 2

Fourier Transform Properties

2.1 Symmetry

To appreciate the symmetric properties of the Fourier transform, we must understand the decomposition of functions into even and odd components. A real function $f(t)$ has a unique decomposition into even and odd functions f_e and f_o , where

$$f_e(t) = \frac{f(t) + f(-t)}{2} \quad (2.1)$$

and

$$f_o(t) = \frac{f(t) - f(-t)}{2} \quad (2.2)$$

2.1.1 Real Functions

If a function $f(t)$ is real-valued, its Fourier transform $F(\omega)$ is Hermitian, or conjugate symmetric. The converse is also true.

$$F(-\omega) = F^*(\omega) \quad (2.3)$$

The Fourier transform decomposes $f(t)$ into a spectrum of complex exponentials $e^{j\omega t}$. Recall that $e^{j\omega t} = \cos(\omega t) + j\sin(\omega t)$. For the integral sum of these exponentials to result in a real function $f(t)$, the real part of $F(\omega)$ must be symmetric about $\omega = 0$ and the imaginary part of $F(\omega)$ must be antisymmetric about $\omega = 0$.

2.1.2 Imaginary Functions

If a function $f(t)$ is imaginary-valued, its Fourier transform $F(\omega)$ is anti-Hermitian, or conjugate antisymmetric. The converse is also true.

$$F(-\omega) = -F^*(\omega) \tag{2.4}$$